

★★★ FINAL EXAM — STARRED PROBLEM TYPES ★★★
△ "NEVER USE WEBERS" — keep magnetic flux as T·m², NOT Wb (1 Wb = 1 T·m²)

① Insulating Sphere — E inside/outside

► Uniform Q in sphere radius R. **Inside (r<R):** E = kQr/R³
Outside (r≥R): E = kQ/r²

Worked: E=760 N/C at r=R/4 (inside): kQ(R/4)/R³ = kQ/(4R²) = 760 → kQ/R² = 3040.
At r=4R (outside): E = kQ/(4R)² = kQ/(16R²) = 3040/16 = **190 N/C**
If asked at r=R: both formulas give E=kQ/R²

② Equivalent Resistance

► **Series:** R_{eq} = R₁+R₂+R₃...
► **Parallel:** 1/R_{eq} = 1/R₁+1/R₂+...
► **Two parallel:** R_{eq} = R₁R₂/(R₁+R₂)

► **Method:** Reduce innermost groups first → simplify outward → I_{total} = ε/R_{eq} → back-substitute V=IR for each branch.

► **Same R in parallel** (n of them): R_{eq} = R/n

③ RC Circuit (timing)

► **Charging:** q(t) = Q_{max}[1-e^{-t/RC}], Q_{max}=Cε
i(t) = (ε/R)e^{-t/RC}
V_C(t) = ε(1-e^{-t/RC}); V_R = εe^{-t/RC}

► **At t=0:** V_C=0, V_R=ε(MAX), i=ε/R(MAX), q=0
At t=∞: V_C=ε(MAX), V_R=0, i=0, q=Q_{max}
τ = RC (63% of final)

► **Discharging:** q=Q₀e^{-t/RC}; i=(Q₀/RC)e^{-t/RC}; V=(Q₀/C)e^{-t/RC}
Correction: exponent is -t/RC (not -ε/RC)

④ B from Moving Charge

► **B = (μ₀/4π)(qv×r̂)/r²**
|B| = (μ₀/4π)(qv sinθ)/r²
► θ = angle between v and r. RHR for direction.

⑤ Long Straight Wire

► **B = μ₀I/(2πr)** (outside, dist r from axis)
► **Direction:** RHR — thumb along I, fingers curl in B direction
► Inside (r<R): B = μ₀Ir/(2πR²)

⑥ Ampere's Law (Multiple Currents)

► **∮B·dl = μ₀I_{encl}**
► **Sign rule:** Curl right hand along path direction. Thumb = + current direction.
★ **CCW path** → + out of page
★ **CW path** → + into page
► **I_{encl}** = sum with signs: I₁(±) + I₂(±) + I₃(±)
► **Then:** B(2πr) = μ₀I_{encl} → B = μ₀I_{encl}/(2πr)

⑦ Toroid B-field

► **B = μ₀NI/(2πr)** (inside, r = dist from CENTER axis)
► N = total turns. Outside: B = 0

△ Flux Φ_B = BA in T·m², not Wb

⑧ Faraday's Law of Induction

► **ε = -dΦ_B/dt** Φ_B = BA cosθ
► **If B(t) varies, θ const:** ε = -A cosθ·(dB/dt)
► **If θ(t) varies, B const:** ε = BA sinθ·(dθ/dt)
► **Both vary:** ε = -A[cosθ·dB/dt - B sinθ·dθ/dt]
► **Rotating coil:** θ=ωt → ε = NBAω sin(ωt); ε_{max} = NBAω
► **N loops:** multiply ε by N

⑨ Slide-Wire Generator

► **ε = vBL** (motional EMF)
► v = rod speed, B = field, L = rod length perp to v & B
► **Induced current:** I = ε/R = vBL/R
Force to push: F = BIL = B²L²v/R
Power: P = I²R = (vBL)²/R

⑩ RL Circuit

► **Energizing:** i(t) = I_{max}[1 - e^{-Rt/L}]; I_{max}=ε/R
Decay: i(t) = I₀e^{-Rt/L}
► **τ = L/R** | V_L(t) = ε·e^{-Rt/L} (energizing)
► **At t=0:** i=0 (inductor blocks). t=∞: i=ε/R

⑪ EM Waves — Full Set

► **E(x,t) = E_m sin(kx - ωt)**
B(x,t) = B_m sin(kx - ωt)
► **Amplitude ratio:** E_m/B_m = c
Instant ratio: E(t)/B(t) = c
► **Wave speed:** c = 1/√(μ₀ε₀) = 3×10⁸ m/s
Wave number: k = 2π/λ [rad/m]
Angular freq: ω = 2π/T = 2πf [rad/s]
c = fλ = ω/k
► **Poynting:** S = (1/μ₀)(E × B) [W/m²]
|S| = EB/μ₀ = E²/(cμ₀) = cB²/μ₀
Avg I: I = E_m·B_m/(2μ₀) = ½cε₀E_m²
► **Direction:** E × B = propagation (RHR)

CH 21-24 CONDENSED (1 Q)

Coulomb & E-fields

F = kq₁q₂/r²; **E = kQ/r²** (point)
Inf line: E=λ/(2πε₀r)=2kλ/r
Inf sheet: E=σ/(2ε₀); **Conductor:** E=σ/ε₀

Ring axis: E=kqz/(z²+a²)^{3/2}
★ Square ctr alt±q: E=4v/2·kg/a²
★ Semicircle ½±q: E=4kQ/(πa²)
★ Zero-E pt (2 like q): x=s/(1+v(q₂/q₁))
★ Two ∥ sheets same sign: between=|σ_A-σ_B|(2ε₀); outside=(σ_A+σ_B)/(2ε₀). Opp sign: swap.

Dipole

τ=p×E; |τ|=pE sinθ
U=-p·E=-pE cosθ; F_{net}=0 unif

Gauss

Φ=∫E·dA=Q_{encl}/ε₀
Inside unif sphere: E=kQr/R³=pr/(3ε₀)
Inside unif cyl: E=pr/(2ε₀)
★ Cavity in sphere: E=ρh/(3ε₀) UNIFORM

Potential

V=kQ/r; U=qV; ΔV=-∫E·dl; E=-dV/dx
W_{field}=qΔV=-ΔU; ½mv²=qΔV (rest)
Plates: E=V/d. Line: ΔV=2kλ ln(r₀/r)

Capacitors

Q=CV; C=ε₀A/d (∥-plate); κε₀A/d w/diel
Series: 1/C_{eq}=Σ1/C_i; Par: C_{eq}=ΣC_i
U=½CV²=Q²/(2C)=½QV; u=½ε₀E²
★ Disconn (Q const): U_{new}/U_{old}=d_{new}/d_{old}
Connected (V const): U_{new}/U_{old}=d_{old}/d_{new}

CH 25 — EMF & CIRCUITS

I=ε/(R+r); V_{term}=ε-Ir (discharge)
V=IR; P=IV=I²R=V²/R; W=εIt
★ Aiding batt: I=(ε₁+ε₂)/ΣR; Opp: I=|ε₁-ε₂|/ΣR (dir = bigger ε wins)

Current Density & Resistivity

I = nqv_dA (n=charge/vol, v_d=drift)
J = I/A = nqv_d [A/m²]
R = ρL/A; ρ=resistivity [Ω·m]; σ=1/ρ
E = ρJ (microscopic Ohm's)
Temp: ρ(T)=ρ₀[1+α(T-T₀)]

CH 26 — KIRCHHOFF & RC

Loop: ΣV=0; Junction: ΣI_{in}=ΣI_{out}
R-series: R_{eq}=ΣR_i; **R-par:** 1/R_{eq}=Σ1/R_i
P_{dissipated} by R = I²R = V²/R

★ Find Time Algebra (RC/RL)

★ **Charging V_C=αε:** t = -RC·ln(1-α)
α=½ → t=RC·ln(2)≈0.693RC
★ **Decay to fraction f of I₀:** t = -τ·ln(f) (works RC and RL with τ=RC or L/R)
★ **RL energize to fraction f of I_{max}:** t = -(L/R)·ln(1-f)
★ **τ table:** 1τ→63%, 2τ→86%, 3τ→95%, 5τ→99.3%

★ Energy-Storage Cheat

Element	Energy	Density
Capacitor	½CV ² =Q ² /2C	u=½ε ₀ E ²
Inductor	½LI ²	u=B ² /(2μ ₀)
EM Wave	—	u=ε ₀ E ² =B ² /μ ₀

In EM wave: u_E = u_B (equal split)

★ Math Quick-Ref

Derivatives: d/dt(e^{at})=ae^{at}; d/dt(sin ωt)=ω cos ωt; d/dt(cos ωt)=-ω sin ωt
Vectors: A·B=AB cosθ (scalar); A×B=AB sinθ (vector ⊥ both, RHR)

Geometry: Circle area=πr²; Sphere area=4πr², vol=(4/3)πr³; Cylinder side=2πrL; Annulus=π(R₂²-R₁²)

Quadratic: x = (-b±√(b²-4ac))/(2a)

Logs: ln(e^x)=x; ln(ab)=ln a+ln b; ln(a/b)=ln a-ln b; ln(1)=0; ln(2)≈0.693

Trig: sin²+cos²=1; sin(2θ)=2sinθcosθ; cos(2θ)=cos²θ-sin²θ
Common angles: sin/cos at 0°=0/1; 30°=½/(√3/2); 45°=√2/2; 60°=(√3/2)/½; 90°=1/0

CONSTANTS & CONVERSIONS

k = 1/(4πε₀) = **8.99×10⁹ N·m²/C²**
ε₀ = **8.854×10⁻¹² C²/N·m²**
μ₀ = **4π×10⁻⁷ = 1.257×10⁻⁶ T·m/A**
μ₀/(4π) = **10⁻⁷ T·m/A**
c = 1/√(μ₀ε₀) = **2.998×10⁸ m/s**
e = **1.602×10⁻¹⁹ C**
m_e = **9.109×10⁻³¹ kg** = 0.511 MeV/c²
m_p = **1.673×10⁻²⁷ kg** = 938 MeV/c²
g = **9.81 m/s²**
1 eV = **1.602×10⁻¹⁹ J**
1 N/C = **1 V/m**
1 Wb = **1 T·m²** △ DON'T USE Wb
1 H = **1 V·s/A = 1 T·m²/A**
1 Hz = **1/s**; 1 rad ≈ 57.3°
1 G = **10⁻⁴ T**; 1 mT = **10⁻³ T**

△ UNITS REFERENCE								
Quantity	Sym	Unit						
Charge	q, Q	C = A·s						
E-field	E	N/C = V/m						
Potential	V	V = J/C						
Energy	U	J = C·V						
Capacitance	C	F = C/V						
Current	I	A = C/s						
Curr density	J	A/m ²						
Resistance	R	Ω = V/A						
Resistivity	ρ	Ω·m						
Power	P	W = J/s						
EMF	ε	V						
B-field	B	T = N/(A·m)						
Mag flux Δ	Φ, B	T·m ² (NOT Wb)						
Inductance	L	H = V·s/A						
Reactance	X _L , X _C	Ω						
Impedance	Z	Ω						
Mag moment	μ	A·m ² = J/T						
Frequency	f	Hz = 1/s						
Angular freq	ω	rad/s						
Wavelength	λ	m						
Wave number	k	rad/m						
Period	T	s						
Energy dens	u	J/m ³						
Intensity	I, S	W/m ²						
Rad pressure	P _{rad}	Pa = N/m ²						
Linear chg	λ	C/m						
Surface chg	σ	C/m ²						
Volume chg	ρ	C/m ³						
Permittivity	ε ₀	F/m						
Permeability	μ ₀	H/m						
Dipole moment	p	C·m						
Drift vel	v _d	m/s						
Conductivity	σ	S/m						
SI	p	n	μ	m	c	k	M	G
10 [^]	-12	-9	-6	-3	-2	3	6	9

★ WORKED PRACTICE EXAMPLES & DECISION TREE

WE-1 ► RC Charging

R=10 kΩ, C=100 μF, ε=12 V. Find t for V_C=8 V.
τ = RC = **1.0 s**; **1.12 s** (1-e^{-t/τ}) → e^{-t/τ} = 1/3
t = -τ ln(1/3) = **8.10 s**

WE-2 ► Long Wire B

I=5 A, r=2 cm.
B = μ₀I/(2πr) = (10⁻⁷·5)/(0.02) = **5×10⁻⁵ T**
Sanity: ≈ Earth's B (50 μT) ✓

WE-3 ► Mass Spec

V=1000 V, B=0.5 T, R=0.05 m.
m/q = B²R²/(2V) = (0.25)/(0.0025)/2000 = **3.13×10⁻⁷ kg/C**
Singly-ionized: m ≈ **5×10⁻²⁶ kg** (~30 amu)

WE-4 ► Slide-Wire

B=0.4 T, L=0.3 m, v=2 m/s, R=5 Ω.
ε = BLv = **0.24 V**; I = **48 mA**
F_{push} = B²L²v/R = **5.76 mN**; P = εI = **11.5 mW**

WE-5 ► Rotating Coil

N=50, A=0.01 m², B=0.2 T, ω=377 rad/s.
ε_{max} = NBAω = **37.7 V (peak)**
V_{rms} = 37.7/√2 = **26.7 V**

WE-6 ► Inductor Energize

ε=24 V, R=12 Ω, L=0.6 H. Find i at t=0.05 s.
τ = L/R = 0.05 s (one tau!); I_{max} = 2 A
i(t) = I_{max}(1-e^{-t/τ}) = **1.26 A**

WE-7 ► LRC Resonance + EM Wave

L=0.1 H, C=10 μF; ω₀ = 1/√(LC) = **1000 rad/s**;
f₀ = **159 Hz**
EM wave E_m=100 V/m; B_m = E/c = **3.33×10⁻⁷ T**; I = ½cε₀E_m² = **13.3 W/m²**

★ COMMON PRODUCTS

μ₀/(4π) = 10⁻⁷ T·m/A | Z₀ = 1/(cε₀) ≈ **377 Ω**
1/√2 ≈ 0.707; e⁻¹ ≈ 0.368; π² ≈ 9.87
ln(2) ≈ 0.693; ln(10) ≈ 2.303; ln(3) ≈ 1.099

★ DECISION TREE — Pick Formula Fast

Charge in B-field, find r → r = mv/(qB)

Two parallel wires → F/L = μ₀I₁I₂/(2πd)

Loop moves through B → ε = BLv

B(t) changing, loop fixed → ε = -A(dB/dt)

Loop rotating in const B → ε = NBAω sin(ωt)

Inductor at t=0: i=0 (open) | at t=∞: i=ε/R

Capacitor at t=0: V=0 (wire) | at t=∞: i=0

AC find Z → √(R²+(X_L-X_C)²)

AC resonance ω → 1/√(LC)

EM wave: E given, find B → B = E/c

EM wave intensity → ½cε₀E_m²

Cap connected, d↑ → V const, U↓

Cap disconnected, d↑ → Q const, U↑

Wire R from material → R = ρL/A

Power dissipated → I²R = V²/R